AI-generated · Unofficial practice

Physics Mock

Answers and Marking Scheme

M: method mark A: answer mark C: communication / concept mark. For revision only; other valid methods should also be credited.

Section A Astronomy and Space Science

1	2	3	4	5	6	7	8
B	C	D	A	C	B	A	D

1.1. B With a speed slightly below the circular-orbit speed, the satellite enters an elliptical orbit with P as the farthest point from the Earth (where its speed is least); the rest of the orbit lies inside the original circle: path B. It cannot move in a smaller circle at once because it is still at P.

1.2. C $T^2 \propto r^3$: $T_Y = 2.0 \times (2.25)^{3/2} = 2.0 \times 3.375 = 6.75$ days.

1.3. D If the Earth orbited the Sun, stellar parallax should be seen; its absence was used to argue that the Earth is stationary: (1) correct. Kepler's first law: (2) correct. Copernicus still used circular orbits (with epicycles), so his predictions were about as accurate as Ptolemy's: (3) wrong.

1.4. A Smaller magnitude means brighter: X appears brighter, so B is wrong; $M_X < M_Y$ so X is more luminous, D is wrong. $m - M = 5 \log \dfrac{d}{10}$: for X it is 5, $d = 100$ pc; for Y it is 3, $d = 39.8$ pc. X is farther: A correct, C wrong.

1.5. C $d = \dfrac{1}{p}$: $\dfrac{d_Q}{d_P} = \dfrac{0.050}{0.020} = 2.5$. $L = 4\pi d^2 b$: $\dfrac{L_Q}{L_P} = \dfrac{1}{5} \times 2.5^2 = 1.25$.

1.6. B Hubble's law holds for an observer in any galaxy. P and Q are 200 Mpc apart, so $v = 70 \times 200 = 1.4 \times 10^4$ km s^{-1}: (1) correct. From Q every galaxy recedes: (2) correct. From P, M also recedes: (3) wrong.

1.7. A Absorption lines identify elements in the outer layers: (1) correct. The peak wavelength of the continuous spectrum (Wien's law) gives the surface temperature: (2) correct. A spectrum (including its Doppler shift) gives composition, temperature and radial velocity, but not the distance by itself: (3) wrong.

1.8. D $v = c\dfrac{\Delta\lambda}{\lambda} = 3.00 \times 10^8 \times \dfrac{6.6}{656.3} = 3.0 \times 10^6$ m s^{-1}, a redshift so receding: (1) correct.
$d = \dfrac{v}{H_0} = \dfrac{3.0 \times 10^3}{70} = 43$ Mpc: (2) correct. $\Delta\lambda \propto \lambda$: H$\beta$ shifts by $486.1 \times \dfrac{6.6}{656.3} = 4.9$ nm: (3) wrong.

Structured Question

1. (10 marks)

(a)(i) (1)

A1 $r(\text{AU}) = \theta('') \times d(\text{pc}) = 0.0060 \times 1500 = 9.0$ AU

(a)(ii) (2)

M1 $T = 6.0$ years; $T^2 = \dfrac{r^3}{M}$ (years, AU, solar masses)

A1 $M = \dfrac{9.0^3}{6.0^2} = 20.3$ solar masses

(b)(i) (2)

A1 $\dfrac{45 + (-25)}{2} = 10$ km s^{-1}, receding

A1 $\dfrac{45 - (-25)}{2} = 35$ km s^{-1}

(b)(ii) (2)

A1 $v = \dfrac{2\pi(9.0 \times 1.50 \times 10^{11})}{6.0 \times 3.16 \times 10^{7}} = 4.47 \times 10^{4}\ \text{m s}^{-1}\ (44.7\ \text{km s}^{-1})$

C1 The orbital plane is inclined to the line of sight, so the Doppler shift measures only the component of the velocity along the line of sight.

(c) (3)

C1 An ordinary star of about 20 solar masses would be extremely luminous and easily observed, but X emits no detectable light, so X is not an ordinary star.

C1 Its mass (**20.3** solar masses) is far above the neutron-star limit of about 3 solar masses, so it cannot be a neutron star.

C1 No known pressure can support such a massive compact object against collapse, so it must have collapsed into a black hole.

Section B Atomic World

1	2	3	4	5	6	7	8
A	D	C	B	D	A	C	B

2.1. A Path 1 has the largest deflection and the smallest impact parameter, so it comes closest: (1) correct. KE turns into electric PE on approach, so KE is least at closest approach: (2) correct. Far away all the PE returns as KE, so the speed is unchanged: (3) wrong.

2.2. D In a polycrystalline foil the crystals are randomly oriented, so some planes always satisfy the diffraction condition and the diffracted beams form cones around the axis, giving rings: (1) correct. $\lambda = \dfrac{h}{\sqrt{2meV}} \propto \dfrac{1}{\sqrt{V}}$: 4 times V halves λ, so for small angles the ring radii are roughly halved: (2) correct. At the same speed a neutron is much more massive, so $\lambda = \dfrac{h}{mv}$ is much smaller and the rings are smaller: (3) wrong.

2.3. C $hf = \phi + 1.0\ \text{eV}$ and $1.5hf = \phi + 2.5\ \text{eV}$. Subtracting: $0.5hf = 1.5\ \text{eV}$, so $hf = 3.0\ \text{eV}$ and $\phi = 2.0\ \text{eV}$.

2.4. B $K_{\max} = hc \cdot \dfrac{1}{\lambda} - \phi$: the slope hc is the same for all metals, so the lines are parallel; a larger ϕ gives a larger intercept on the $\dfrac{1}{\lambda}$ axis (shifted right): graph B.

2.5. D $E = \dfrac{hc}{\lambda} = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{1.88 \times 10^{-6}} = 1.06 \times 10^{-19}\ \text{J} = 0.661\ \text{eV}$. $-0.85 - (-1.51) = 0.66\ \text{eV}$, i.e. $n = 4 \to n = 3$.

2.6. A $\lambda = \dfrac{h}{\sqrt{2meV}}$, so equal λ means equal mV: $V_p = \dfrac{150}{1836} = 0.0817\ \text{V}$.

2.7. C $\lambda = \dfrac{h}{\sqrt{2meV}} = \dfrac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(1.0 \times 10^{5})}} = 3.9 \times 10^{-12}\ \text{m}$, far shorter than visible light, giving a much higher resolution: (1) correct. Electrons must pass through the specimen, so it must be very thin: (2) correct. The beam must travel in a vacuum, so living cells cannot be viewed directly: (3) wrong.

2.8. B At the nanoscale gold particles absorb and scatter light differently, so the solution looks red or purple: (1) correct. Number of small cubes $= \left(\dfrac{1000}{10}\right)^3 = 10^6$; total area $= 10^6 \times 6(10)^2 = 6 \times 10^8\ \text{nm}^2$ versus $6(1000)^2 = 6 \times 10^6\ \text{nm}^2$ originally: 100 times: (2) correct. Nanomaterial properties can differ greatly from the bulk: (3) wrong.

Structured Question

2. (10 marks)

(a) (1)

C1 Electrons in the atom can only occupy discrete energy levels, so the photons emitted in transitions have only certain energies $hf = E_i - E_f$, i.e. certain wavelengths.

(b)(i) (1)

A1 3 ($3 \to 2$, $2 \to 1$, $3 \to 1$)

(b)(ii) (2)

A1 Downward arrow from $n = 3$ to $n = 2$.

C1 It has the smallest energy gap, and $\lambda = \dfrac{hc}{\Delta E}$ is largest for the smallest ΔE.

(b)(iii) (3)

M1 $\Delta E = -1.51 - (-3.40) = 1.89\ \text{eV} = 3.02 \times 10^{-19}\ \text{J}$

A1 $\lambda = \dfrac{hc}{\Delta E} = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{3.02 \times 10^{-19}} = 6.58 \times 10^{-7}\ \text{m}$

C1 Visible light (red).

(b)(iv) (3)

M1 Photon energy $= 1.51 + 0.50 = 2.01\ \text{eV} = 3.22 \times 10^{-19}\ \text{J}$

A1 $\lambda = \dfrac{hc}{E} = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{3.22 \times 10^{-19}} = 6.18 \times 10^{-7}\ \text{m}$

C1 No: ionizing from the ground state needs $13.6\ \text{eV} > 2.01\ \text{eV}$.

Section C Energy and Use of Energy

1	2	3	4	5	6	7	8
D	B	B	C	A	C	A	D

3.1. D Luminous flux $= 90 \times 12 = 1080\ \text{lm}$; light power $= \dfrac{1080}{500} = 2.16\ \text{W}$; efficiency $= \dfrac{2.16}{12} = 18.0\%$.

3.2. B $P = \frac{1}{2}\rho A v^3$: doubling v gives 8 times P: (1) correct. $A = \pi r^2$, so doubling the blade length gives 4 times A: (2) correct. $P \propto \rho$: (3) wrong.

3.3. B $P = 800 \times 2.5 \times \cos 30° \times 0.18 = 312\ \text{W}$.

3.4. C Total wall area $= 2(30 + 20) \times 15 = 1500\ \text{m}^2$; windows $= 40 \times 3.75 = 150\ \text{m}^2$; opaque wall $= 1350\ \text{m}^2$.
$\text{OTTV} = \dfrac{15(1350) + 120(150)}{1500} = 25.5\ \text{W m}^{-2}$.

3.5. A The air layer has a low thermal conductivity, so double glazing has a smaller U-value: (1) correct. The U-value is already per unit area, independent of area: (2) wrong. $\dfrac{P}{A} = U\Delta T$: (3) correct.

3.6. C Input power $= \dfrac{2.8}{4.0} = 0.70\ \text{kW}$; operating time $= 150 \times 6 = 900\ \text{h}$; energy $= 0.70 \times 900 = 630\ \text{kWh}$.

3.7. A $v = \sqrt{2gh} = \sqrt{2(9.81)(45)} = 29.7 \approx 30\ \text{m s}^{-1}$: (1) correct.
$P = 0.80 \times 1000 \times 20 \times 9.81 \times 45 = 7.1 \times 10^6\ \text{W}$: (2) correct. The PE released per second, $\rho Q g h$, depends only on Q and h; a narrower pipe does not increase it and only adds friction losses: (3) wrong.

3.8. D Per fission $E = 0.215 \times 931 = 200\ \text{MeV} = 3.20 \times 10^{-11}\ \text{J}$. Thermal power $= \dfrac{900}{0.33} = 2.73 \times 10^3\ \text{MW}$.
Fissions per day $= \dfrac{2.73 \times 10^9 \times 86400}{3.20 \times 10^{-11}} = 7.36 \times 10^{24}$. Mass $= \dfrac{7.36 \times 10^{24}}{6.02 \times 10^{23}} \times 0.239 = 2.92\ \text{kg}$.

Structured Question

3. (10 marks)

(a) (1)

C1 No exhaust gases (e.g. NOx, particulates) are emitted at street level, improving roadside air quality (or: less noise).

(b)(i) (2)

M1 $300\ \text{kWh} = 300 \times 3.6 = 1080\ \text{MJ}$

A1 $\dfrac{1080}{250} = 4.32\ \text{MJ km}^{-1}$

(b)(ii) (1)

A1 $0.35 \times 36 = 12.6\ \text{MJ km}^{-1}$

(c)(i) (2)

M1 $v = 13.9\ \text{m s}^{-1}$; $\text{KE} = \frac{1}{2}(14000)(13.9)^2 = 1.35 \times 10^6\ \text{J}$

A1 $P = \dfrac{1.35 \times 10^6}{12} = 1.13 \times 10^5\ \text{W}$

(c)(ii) (2)

A1 $\dfrac{1.13 \times 10^5}{0.85} = 1.32 \times 10^5\ \text{W}\ (132\ \text{kW})$

C1 Motor output needed is $113\ \text{kW} < 200\ \text{kW}$ (or input $132\ \text{kW}$ is within the motor rating), so the motor can cope.

(d) (2)

M1 Primary energy for electric bus $= \dfrac{4.32}{0.40} = 10.8\ \text{MJ km}^{-1}$

C1 $10.8 < 12.6\ \text{MJ km}^{-1}$: the electric bus uses less primary energy, so the statement is not correct (though the saving is only about 14%).

Section D Medical Physics

1	2	3	4	5	6	7	8
B	A	D	D	B	C	A	C

4.1. B Q is the ossicles, which act as levers passing the vibrations of the eardrum to the much smaller oval window, amplifying the pressure. P is the eardrum, R the semicircular canals (balance), S the cochlea.

4.2. A TIR occurs at the core–cladding boundary, keeping the light in the core; without cladding, light would leak into an adjacent fibre where their surfaces touch: (1) correct. $\sin c = \dfrac{1.40}{1.50}$, $c = 69.0°$: (2) correct. A larger entry angle makes the ray inside the core steeper relative to the axis, so its angle of incidence at the boundary is smaller and less likely to exceed the critical angle: (3) wrong.

4.3. D Depth $= \dfrac{1540 \times 52 \times 10^{-6}}{2} = 0.040\ \text{m} = 4.0\ \text{cm}$: (1) correct. Echo strength shown as peak height belongs to an A-scan; a B-scan uses the brightness of dots: (2) wrong. Gel removes the air layer, avoiding almost total reflection at an air–skin boundary: (3) correct.

4.4. D Ultrasound is non-ionizing and safe for the fetus: (1) correct. It gives real-time images: (2) correct. CT generally has a higher resolution, so (3) is not a reason.

4.5. B $\alpha = \left(\dfrac{7.8 - 1.6}{7.8 + 1.6}\right)^2 = 0.435$. Both boundaries have the same α, so the fraction transmitted $= (1 - 0.435)^2 = 0.319$.

4.6. C $e^{-\mu(4.0)} = 0.60$ gives $\mu = 0.128\ \text{mm}^{-1}$ and half-value thickness $= \dfrac{\ln 2}{\mu} = 5.43\ \text{mm}$. $\dfrac{15\%}{60\%} = \dfrac{1}{4}$, i.e. two more half-value thicknesses: $2 \times 5.43 = 10.9\ \text{mm}$.

4.7. A 24 h is 4 half-lives: $\left(\frac{1}{2}\right)^4 = 6.25\%$: (1) correct. The collimator passes only γ rays travelling along its holes, so the image maps positions in the body: (2) correct. γ rays are weakly ionizing; they are used because they are penetrating enough to leave the body and be detected and do little damage: (3) wrong.

4.8. C Absorbed dose: $\alpha = \dfrac{2.0}{0.50} = 4.0\ \text{mGy}$, $\gamma = \dfrac{4.0}{0.50} = 8.0\ \text{mGy}$. Equivalent dose $= 20(4.0) + 1(8.0) = 88\ \text{mSv}$.

Structured Question

4. (10 marks)

(a)(i) (2)

M1 Object at 25 cm forms a virtual image at 75 cm: $P = \dfrac{1}{0.25} - \dfrac{1}{0.75}$

A1 $P = +2.67\ \text{D}$ (converging)

(a)(ii) (2)

A1 One ray from the top of O parallel to the axis, refracted through F on the far side; one ray through the optical centre undeviated.

A1 Both refracted rays traced back (dashed) to meet on the same side as O, giving an erect, magnified virtual image I about 75 cm from the lens.

(a)(iii) (2)

A1 Farthest object at the focal point: $f = \dfrac{1}{2.67} = 0.375\ \text{m} = 37.5\ \text{cm}$ (image at infinity, her far point).

C1 Objects beyond 37.5 cm form real images that her eye cannot focus, so distant objects look blurred through the glasses; she must view them directly.

(b)(i) (2)

M1 $d = \dfrac{vt}{2} = \dfrac{1540 \times 92 \times 10^{-6}}{2}$

A1 $= 0.0708\ \text{m}\ (7.08\ \text{cm})$

(b)(ii) (1)

A1 $\dfrac{1540 \times (92 - 26) \times 10^{-6}}{2} = 0.0508\ \text{m}\ (5.08\ \text{cm})$

(b)(iii) (1)

C1 Any one: ultrasound is attenuated (absorbed/scattered) over the longer path; part of the energy is reflected at the first boundary so less reaches the second; the second boundary may have a smaller impedance mismatch.